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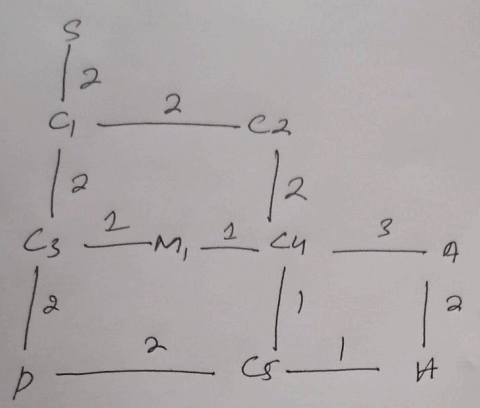
Task 2

Question 1

Question #01

Abdul Kalam (Task 2)
23p-0622
BS-CS-6A

Graph for problem



Start : S
Goal : H

BFS : Queue | S | C1 | C2 | C3 | C4 | M | P | A | C5 |

visted : S, C1, C2, C4, A, H

1) S neighbours → C1

2) C1 neighbours → C3, C2
select C2.

3) C2 neighbours → C4
added to Queue.

4) C3 neighbours → M, P
added to Queue

5) C4 neighbours → A, C5
added to Queue

6) M neighbour C4
already.

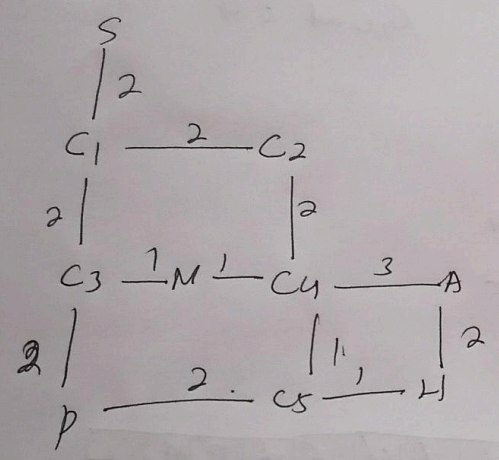
7) P, → C5
already.

8) A neighbour H.
Goal State

path found.

$S \to C_1 \to C_2 \to C_4 \to A \to H.$

Using UCS.



using UCS :

US uses heap to keep track of priority

Cost [2 + 2 + 2 + 1

1) $S \xrightarrow{2} C_1$

2) $C_1 \xrightarrow{2} C_2$

$C_1 \xrightarrow{2} C_3$

added 2.

path $S \to C_1 \to C_2$.

3) $C_2 \xrightarrow{2} C_4$ added 2 $S \to C_1 \to C_2 \to C_4$

4) $C_4 \xrightarrow{3} A$

$C_4 \xrightarrow{1} C_5$

added 1

$S \to C_1 \to C_2 \to C_4 \to C_5$

Question 2

5) $C_5 \xrightarrow{2} P$ added 1 equal

$C_5 \xrightarrow{1} A$

path : $S \to C_1 \to C_2 \to C_4 \to C_5 \to A$

Total cost : 8.

Question #02.

Ride sharing problem:

1) State Space:

- driver current location (L)
- which passenger have been picked up. (P)

State = (L, P)

if n passengers $(n+1) \times 2^n$

2) Actions:

A1 : go to unpick passengers.

move from current location to p_i

Cost : Euclidean distance

3) using A*:

$f(n) = g(n) + h(n)$

priority Que is used to expand with smallest f(n).

Goal Test:

All passengers picked (and destination reached if required).

4) Heuristic function:

h(n) = Distance to nearest unpicked passengers

h(n) = minimum distance to unpick location + minimum spanning tree cost of remaining distance to passengers

5) Is heuristic Admissible?

h(n) ≤ true remaining cost.

- it never overestimates actual cost remaining
it is admissible.

Question 3

Question 3

1) State definition:

- The state is defined as configuration of 3x3 board
- each represents a position of title 1–8 and empty space.

example $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 5 \\ 7 & 2 & 6 \end{bmatrix}$ (0 = _ blank)

state = arrangement of tiles

$= 9!$

$= 362,880$ but only half are reachable from given space.

2) Actions:

possible moves:

- move up, down, left, right

Each move:

- swap blank with adjacent title.
- has cost 1.

step cost = 1

3) A* (with manhattan)

$f(n) = g(n) + h(n)$.

$g(n)$ = number of moves so far
$h(n)$ = heuristic estimate to goal

- Manhattan distance

$d = |x_{current} - x_{goal}| + |y_{current} - y_{goal}|$

e.g 5 tile
current pos (1,2) Goal (1,1)
$|1-1| + |2-1| = 1.$

4) Is manhattan admissible?

Yes, because it never over estimates no of moves, each tile must be move atleast its manhattan distance
$h(n) \leq$ actual cost

5) Nodes explored A^* vs BFS

BFS	A^*
• explore all stages	• uses heuristic guide toward goal.
• doesnt use heuristic	• expand only few nodes (know
Time Complexity $O(b^m)$	• guarantee optimal solution
deep puzzle it explore thousands of states	• much better for bigger problem.
	• less Time Complexity than bfs.